07 May 2021

EM FX: Does the acceleration in commodity prices warrant long EM FX positions?

Not in our view; here's why

- The commodity rally has proven to be underwhelming for EM currencies. The correlation of EM currencies to commodities have been well below their historical norms this year.
- There are multiple reasons for this, including idiosyncratic shocks in the major EM commodity exporters and rising US yields year-to-date.
- The underperformance raises the question of whether there is a catch-up trade to be had in EM FX.
- We attempt to answer three questions in this note: First, can any valuation gap between EM FX and commodities close? Second, is there a case to be long EM FX against the dollar on only continued commodity price momentum? Third, is there a relative value case to be long EM commodity exporters against importers on rising commodity prices?
- We cannot find a convincing case to be broadly long EM commodity currencies against the dollar based on just the commodity price story. Valuation gaps do not typically close, and commodity price momentum from current levels is more likely to taper off. However, we do find more convincing evidence for a relative value argument to be long EM commodity exporters versus importers.
- Medium term, still high political/fiscal risk premia in major commodity exporters is likely to continue to weigh on the beta of EM currencies to commodity prices with the absence of FX carry compensation; the slowing in China's credit growth and fixed asset investment should weigh on its commodity imports and consequently commodity price momentum; and, the potential for higher US real yields complicate the region's currencies as a commodity play.
- We therefore stay generally sidelined in EM commodity currencies. The only exception is our OW position in ZAR against CEE FX; this is consistent with our finding that the "best" case to be long EM commodity currencies is on a relative value basis.

EM FX lags the commodity rally

The commodity rally has proven to be underwhelming for EM currencies. For most of this year we have been surprised by the breadth and magnitude of the rally in commodities prices. Several factors have fueled this, including stronger global growth expectations, metals demand from China, and speculative activity driven by global liquidity conditions. Yet, EM currencies, which traditionally have a high correlation with commodities prices, both because of the large number of EM commodity exporters but also the correlation of commodities prices with risk appetite, have not benefitted as they usually do from the commodity price rally (Exhibit 1). EM FX correlations to commodities are also far lower than the historical average of the past 5-years (Exhibit 2).

Particularly for Latam commodity currencies, the correlation with commodity prices has been close to zero or even negative in recent months. Part of the low correlation with commodity prices this year can be explained by idiosyncratic shocks in several commodity exporters, including Russia, Brazil, Colombia and Peru. Additionally, the rise in commodities prices has coincided with higher US real yields, which has offset some of the tailwinds from higher commodity prices. Nonetheless, the scope of the underperformance of EM currencies raises the question whether investors should enter long positions in EM FX as a catch-up trade to commodities.

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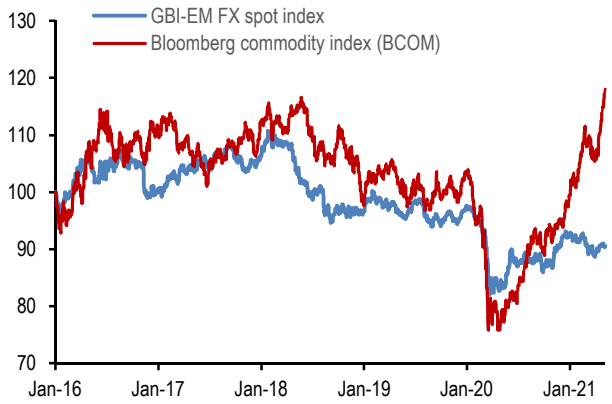
Are rising commodity prices a reason to go long EM FX? We think the answer to that is mostly a 'no', for two main reasons. First, we find that large 'dislocations' between EM currencies and commodities do not close. Second, we are not convinced that there is significant upside in broad commodities from current levels, so the momentum argument is not strong. That said, our analysis suggests there is some scope, historically, for relative value outperformance of commodity currencies. Accordingly, we are generally sidelined in commodity currencies, only maintaining a relative value OW in ZAR in the commodity FX bloc.

Exhibit 1: EM FX has failed fully to benefit from the rally in commodity prices over recent months

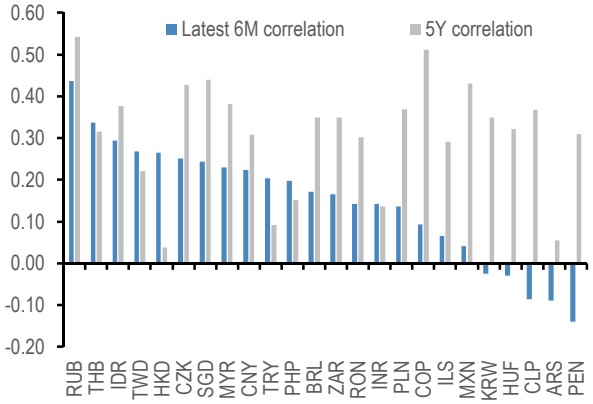
Indexed to 100 as of Jan-16



Source: J.P. Morgan, Bloomberg Finance L.P. Data as of 05/05/21

Exhibit 2: The correlation of EM FX to commodity price moves is lower now relative to history

Correlation coefficients



Source: J.P. Morgan, Bloomberg Finance, L.P.

History suggests 'catch up' to commodities is unlikely

We ask three main questions and see if they have worked historically in out of sample backtests:

- First, do large dislocations in EM currencies against the dollar against commodities close? *Spoiler alert*: no.
- Second, is there a case to be long EM currencies against the dollar if there is strong price appreciation of commodities? *Spoiler alert*: This strategy has mixed performance historically, and one would need to expect further increases.
- Third, is there a case to hold intra-EM relative value positions on rising commodity prices? *Spoiler alert:* We find reasonable evidence that this strategy works historically. While we are not convinced of the prospects from the current high levels of commodity prices, we think, on balance, it is worth having some allocation to this view, which we express via ZAR.

To answer the questions above, we run simple regressions to assess whether EM FX is cheap to commodity prices, and if so, whether valuation mean reversion has worked historically. We also look at simple commodity momentum-based strategies to trade EM FX, both from a time series perspective (using our GBI-EM FX TR basket), but also from a cross-sectional perspective (building a long/short EM FX basket based on terms of trade momentum).

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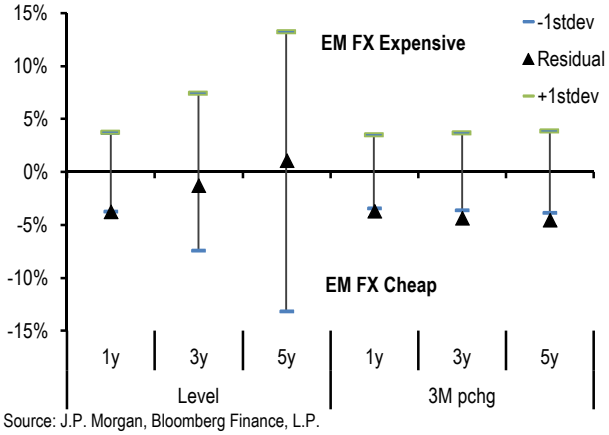
Exhibit 3: EM FX is generally around 1stdev cheap to commodity prices in our regression

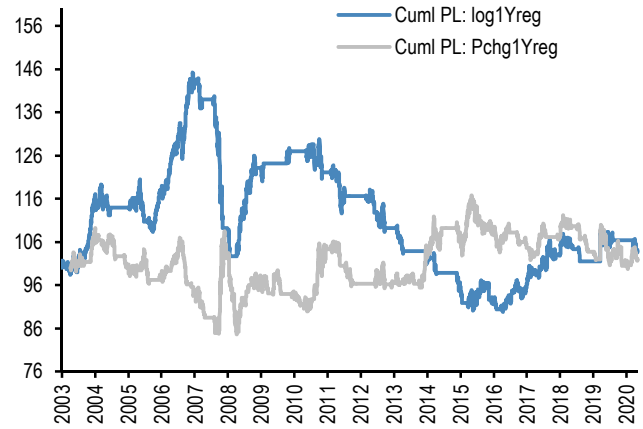
Residuals and Stdev bands across regression horizons



Source: J.P. Morgan, Bloomberg Finance, L.P.

Exhibit 4: Cumulative P/L is generally poor for these mean reversion value strategies

Cumulative P/L since 2003 (ex-transaction costs)

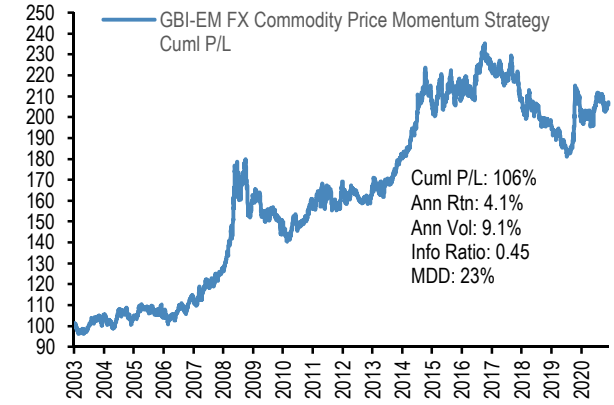


Source: J.P. Morgan, Bloomberg Finance, L.P.

While there is evidence to suggest EM FX is cheap to commodity prices in our regressions, we find that mean reversion valuation backtests have generally performed poorly, particularly post GFC. We perform simple rolling window regressions of GBI-EM EM TR FX on Bloomberg's commodity price index in both log level and three-month percent changes, and across various horizons since 2003. We generally find that EM FX trades around 1 standard deviation cheap to commodity prices (Exhibit 3). However, mean reversion backtests for both level and change regressions perform poorly (Exhibit 4): we buy/sell EM FX vs USD when the residual of the regression breaches -/+ 1 standard deviation and close the trade when the residual moves to zero (we exclude transaction costs). The results do not change much when we adjust the standard deviation entry threshold. Reassuringly, results are similar when we substitute commodity prices for aggregate EM terms of trade. This analysis makes it hard to make the case for an EM FX mean reversion catch-up trade to commodity prices.

Exhibit 5: Cuml P/L for trading GBI-EM FX on a simple 3M Z-score commodity price momentum strategy since 2003

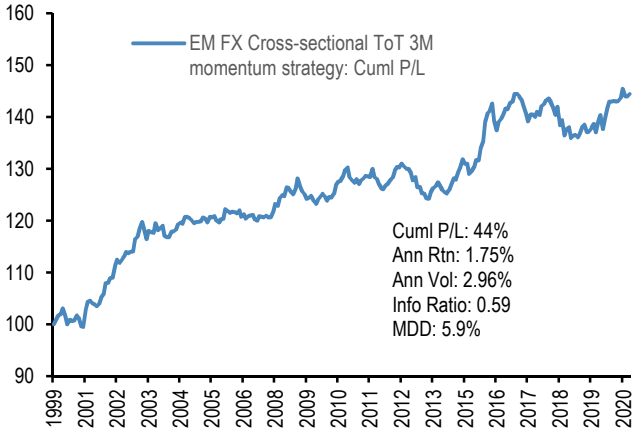
Cumulative P/L (ex-transaction costs)



Source: J.P. Morgan, Bloomberg Finance, L.P.

Exhibit 6: Cumulative P/L for an EM FX cross-sectional strategy on EM 3M ToT momentum since 2000

Cuml P/L: Long top vs bottom 25% ToT momentum (ex-transaction costs)



Source: J.P. Morgan, Bloomberg Finance, L.P.

EM FX commodity price momentum strategies have performed better than mean reversion strategies with time series momentum (and cross-sectional) strategies recommending long EM FX (commodity) positions. As well as mean reversion strategies, we also explore momentum-based commodity price strategies for EM FX. For our time series momentum strategy, we take a 3-month z-score of Bloomberg's commodity price index, buying EM FX when it is positive and selling it when it is negative. The back test results perform better than the mean reversion strategies (Exhibit 5), though with large drawdowns. Currently, the strategy is recommending being long EM FX vs USD.

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Cross-sectional strategies work best. We also look at momentum on a cross-sectional EM basis, using a 3-month z-score of individual country commodity terms of trade using IMF data (lagged by 3months due to data availability), buying the top 25% with the highest terms of trade momentum and selling the bottom 25% with the lowest momentum. The strategy has performed respectably over time with manageable drawdowns (Exhibit 6), and unsurprisingly is currently recommending long positioning in commodity exporters (IDR, MYR, RUB, COP, BRL, PEN) vs importers (KRW, CNY, SGD, THB, CZK, TRY) given the positive momentum in commodity terms of trade. This analysis suggests there is a stronger case for being long commodity EM FX on a momentum argument rather than a valuations/mean reversion basis.

We see good reasons based on risk-reward to override the long EM FX signals from the commodity momentum strategies. While we think there could a short window in which EM FX could benefit from the commodity price momentum due to stable US real yields and EUR/USD, we think there are several reasons which suggest that there is a poor risk reward for both chasing commodity prices at this point but also expressing a commodity trade via EM FX. We expand on these factors below:

1) **China's credit growth and fixed asset investment is slowing and will likely weigh on commodity price momentum in our view.** Our China economists continue to call for a rotation from investment to consumption/services growth in the context of a slowing in fiscal and credit support (see here and here G.Ng, H.Zhu). This should lead to a slowing in growth momentum in the credit intensive sectors such as infrastructure and real estate, which will consequently lead to a reduced demand in China's commodity imports, particularly of metals prices (Exhibit 7 and Exhibit 8). Moreover, our commodity team does not believe the green demand for metals (such as copper) will be strong enough yet to offset this slowing in Chinese demand (see here). We also see steel product inventories past their pre-pandemic averages, potentially signaling less new demand ahead. All of these factors we think reduces the risk-reward around FX-based commodity momentum strategies.

Exhibit 7: The forecasted slowing in China's fixed asset investment should weigh on their commodity imports

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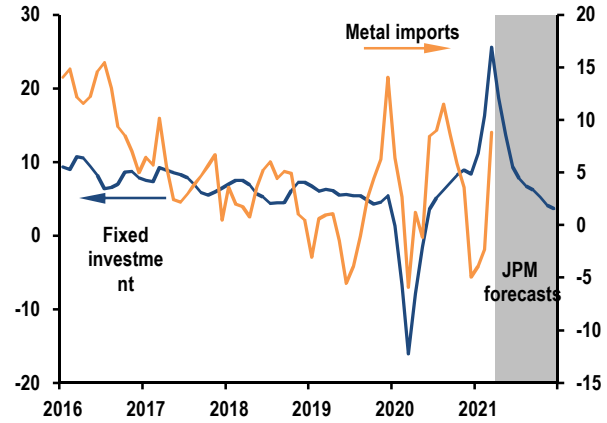
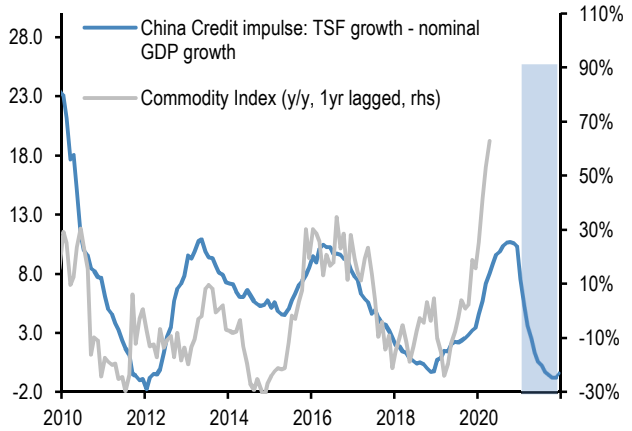


Source: NBS, China Customs, J.P. Morgan

Exhibit 8: China's credit impulse trajectory suggests commodity prices momentum will slow

Cumulative P/L since 2003 (ex-transaction costs)



Note: Blue area indicates JPM TSF y/y forecast; Source: J.P. Morgan, Bloomberg Finance, L.P.

2) **Lower correlation of EM FX to commodity prices might be associated with the increase in political and fiscal risks.** The correlation between EM FX and commodity prices has fallen in recent months relative to its medium-term correlation. This can be explained by two factors: 1) Historically low EM FX carry, making it less compelling; 2) the rise in the region's political and fiscal risk premia relative. These two factors together suggest that the region is offering less yield for higher risks, resulting in a lower beta to commodity prices (**Error! Reference source not found.**). With the exception of Brazil (where we see some marginal improvement in fiscal/political risk premia – see here), we don't see this changing soon for the likes of Colombia (fiscal reform debate), Peru (presidential elections), Chile (heavy political calendar) and Russia. Even South Africa, where we are long, as longer term fiscal risks which, while not immediately acute, are formidable nonetheless. This

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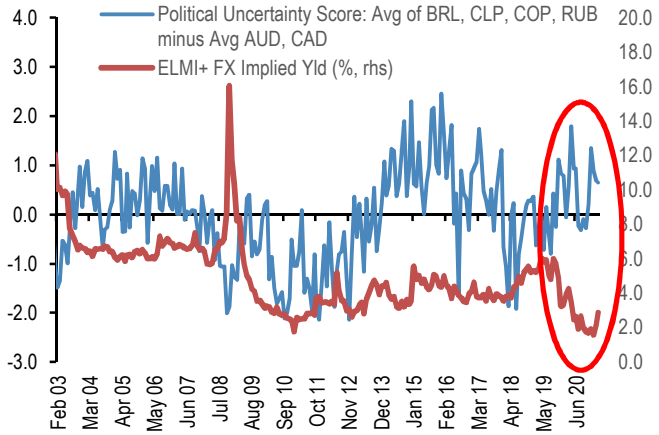
suggests to us that EM's beta to commodity prices may stay more muted, reducing the risk-reward for increasing EM FX longs.

Exhibit 9: EM FX political uncertainty has moved higher as FX implied yield compensation has moved to record lows….

Political uncertainty score: Avg(BRL, CLP, COP, RUB) minus Avg(AUD, CAD) vs EM FX Implied yld



Source: J.P. Morgan, Haver Analytics, Bloomberg Finance, L.P.

Rising US yields are a headwind

A further complication for expressing a commodities view via EM FX is the risk of rising US real yields. Another reason why EM FX has lagged the commodity price rally may be due to the increase in US real yields in Q1, which may have weighed further on EM FX's beta to commodity prices. We have since seen a significant retracement of US real yields from the March highs, with the risk-reward now heavily skewed to further upside potential than downside (Exhibit 10). This threat of an abrupt move higher in US real yields is an additional risk factor that further reduces the risk-reward around using EM FX as a proxy for a commodity momentum trade.

To illustrate this further, we re-run our rolling time series regressions of GBI EM FX on commodity prices to also include US real yields (all in 3Mchgs and across our regression different horizons). Using the sensitivities of the regressions, we calculate how much commodity prices would have to increase from here to offset a 30bp increase in US 10-year real yields (ie back to the top of the YTD range) on EM FX performance. We show that commodity prices would have to increase as much as 6.8% in our 1-year regression in order to keep EM FX stable in the context of a 30bp increase in US real yields (Exhibit 11). This further highlights to us that the bar for further commodity price rises feeding into EM FX gains is only moving higher from here given the risks to higher US real yields.

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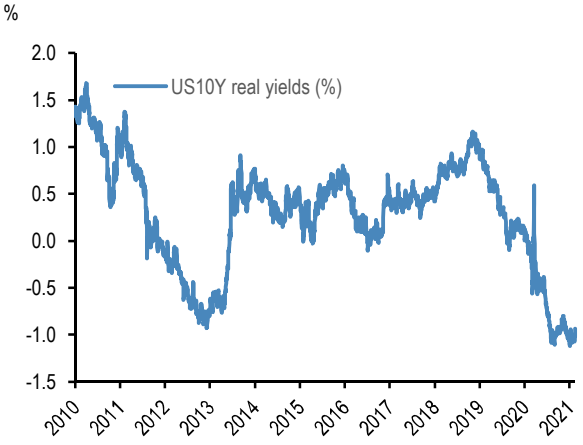
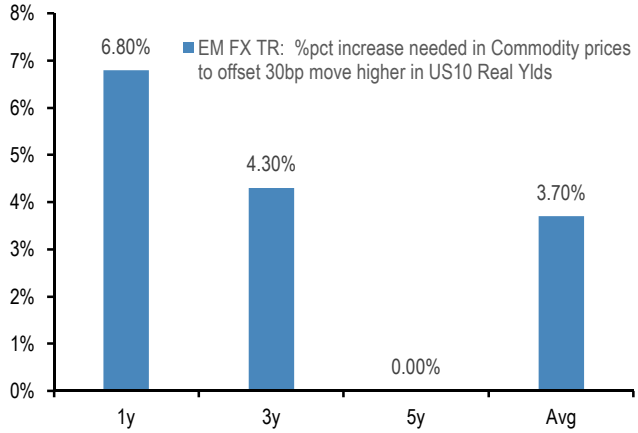
Exhibit 10: How much lower can 10Y US real yields go? The risk reward is skewed to upside from here



Source: J.P. Morgan, Bloomberg Finance, L.P.

Exhibit 11: Commodity prices will have to move increasingly higher to benefit EM FX currencies if US real yields begin to rise again



Source: J.P. Morgan, Bloomberg Finance, L.P.

Conclusion

Our conclusion is that the risk-reward is insufficiently appealing to increase EM FX positions on just a commodity price argument. We come to the conclusion that, while there is evidence that EM FX trades on the cheaper side to commodity prices, mean reversion strategies of this sort simply have not worked well historically, making it difficult to argue for a catch-up trade to commodities. Our momentum strategies, however, fare better with both time series and cross-sectional strategies recommending long EM FX commodity positions. While we see a short-term window for this commodity momentum trade to play out in EM FX, we think the risk-reward of broadly chasing it here is poor given high idiosyncratic risks and low FX carry in commodity exporters; a continued slowing in China credit growth and fixed asset investment which should weigh on China's commodity imports; and, the potential for a retracement higher in US real yields, which complicates the outlook for EM FX. Accordingly, for now, we only stay OW ZAR vs CEE in the GBI EM Model Portfolio.

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